

# Online Appendix to Government Debt Management: The Long and the Short of It\*

Faraglia, Elisa <sup>†</sup>    Marcet, Albert<sup>‡</sup>    Oikonomou, Rigas <sup>§</sup>    Scott, Andrew <sup>¶</sup>

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<sup>†</sup>University of Cambridge and CEPR.

<sup>‡</sup>Institut d'Anàlisi Econòmica-CSIC, ICREA, Barcelona GSE, UAB, MOVE and CEPR.

<sup>§</sup>Université Catholique de Louvain. Email: Rigas.Oikonomou@uclouvain.be

<sup>¶</sup>London Business School and CEPR.

This appendix contains two main sections. Section A contains some analytic results that have been left out of the text. We first give some additional details about the complete markets model. Then we set up the Lagrangeans that have not been included in the main text, namely for the model with coupons of section 7.1, the ‘callable bonds’ model of section 7.3, and the optimal repurchases with transaction costs model of section 6.3. We derive the state vectors and first order optimality conditions for these models.

Section B contains the Numerical Appendix. It discusses in detail the implementation of the ‘Forward States’ and ‘Condensed PEA’ algorithms, several practical issues on solving portfolio models with incomplete markets with the PEA and we report on the accuracy of the simulated model.

## A Extensions and Additional Theoretical Results

### A.1 Complete Markets and No Buyback

We describe in this section the debt management strategy under a complete financial market assuming no buyback. This provides more details for the calculations in section 3.4.1 in the main text.

As in Chari and Kehoe (1999) we formulate the planner’s program as a maximization of the household’s utility subject to the following implementability constraint at date 0:

$$E_0 \sum_{t=0}^{\infty} \beta^t [(u_{c,t} - v_{x,t})(c_t + g_t) - u_{c,t}g_t] = E_0 \sum_{i \in \{1, N\}} \sum_{t=0}^{i-1} \beta^t u_{c,t} b_{t-i}^i$$

where for simplicity we assume that the government may issue debt in one year and  $N$  year maturities.

The optimal allocation derives as a solution to the following equation:

$$(1) \quad \mathcal{L}_{CM} = E_0 \left\{ \sum_{t=0}^{\infty} [\beta^t u(c_t) + v(T - c_t - g_t)] + \Lambda \left[ \sum_{t=0}^{\infty} \beta^t ((u_{c,t} - v_{x,t})(c_t + g_t) - u_{c,t}g_t) - \sum_{i \in \{1, N\}} \sum_{t=0}^{i-1} \beta^t u_{c,t} b_{t-i}^i \right] \right\}$$

The first order conditions for the optimum are given by:

$$(2) \quad u_{c,t} - v_{x,t} - \Lambda[u_{cc,t}c_t + u_{c,t} - v_{x,t} + v_{xx,t}(c_t + g_t)] = 0 \quad \text{for } t > N - 1$$

$$(3) \quad u_{c,t} - v_{x,t} + \Lambda[u_{cc,t}c_t + u_{c,t} - v_{x,t} + v_{xx,t}(c_t + g_t)] - \Lambda u_{cc,0} \sum_{i \in \{1, N\}} b_{t-i}^i = 0 \quad \text{for } t \leq N - 1$$

#### A.1.1 Optimal Debt Management under Buyback

Let  $g^{t-1} = (g_0, g_1, \dots, g_{t-1})$  be the history of government spending shocks up to date  $t - 1$ . As ABN and Faraglia et al. (2010) we define  $z$  as the present discounted value of the government surplus contingent on  $g^{t-1}$  and the current realization of spending  $g_t$ .

$$(4) \quad z_t(g^{t-1}, g_t) = E_t \sum_{i=0}^{\infty} \frac{\beta^i}{u_{c,t}} [(u_{c,t+i} - v_{x,t+i})(c_{t+i} + g_{t+i}) - g_{t+i}u_{c,t+i}]$$

Assuming that debt is not state contingent, and that the asset returns are not perfectly correlated, one of the key results of Buera and Nicolini (2004) is that  $z$  is the only variable (along with the asset prices) that is needed to pin down the optimal issuance of the government. We assume for simplicity that government expenditure follows a two step Markov process taking values  $g_H > g_L$  with probabilities  $\pi_{HH}$  and  $\pi_{LL}$ , of remaining in the same state. After  $t \geq N$  we have the first order conditions above are time-invariant so that  $z_t(g^{t-1}, g_i) = z^i$  and  $p^{N-1}(g^{t-1}, g_i) = p_i^{N-1}$  for  $i = H, L$  and for all  $t \geq N$ .

Clearly surpluses are higher when  $g^L$  occurs, therefore  $z_H < z_L$ .

The optimal portfolio must be such that the outstanding value of the government's liability is equal to  $z^H$  whenever government spending is high and equal to  $z_L$  when spending is low. Recall that under buyback the optimal portfolio needs to satisfy:

$$(5) \quad b_{t-1}^1(g^{t-1}) + p_i^{N-1}(g^{t-1}) b_{t-1}^N(g^{t-1}) = z^i \quad \text{for } i = H, L \quad \forall t.$$

If  $u$  is strictly concave  $p_H^{N-1} \neq p_L^{N-1}$  and we have

$$(6) \quad \begin{pmatrix} 1 & p_H^{N-1} \\ 1 & p_L^{N-1} \end{pmatrix} \begin{pmatrix} b_{t-1}^1(g^{t-1}) \\ b_{t-1}^N(g^{t-1}) \end{pmatrix} = \begin{pmatrix} z^H \\ z^L \end{pmatrix}.$$

yielding

$$(7) \quad \begin{pmatrix} b_{t-1}^1(g^{t-1}) \\ b_{t-1}^N(g^{t-1}) \end{pmatrix} = \begin{pmatrix} \frac{p_H^{N-1} z^L - p_L^{N-1} z^H}{p_H^{N-1} - p_L^{N-1}} \\ \frac{z^H - z^L}{p_H^{N-1} - p_L^{N-1}} \end{pmatrix} = \begin{pmatrix} B^1 \\ B^N \end{pmatrix}$$

The key properties of the solution are that first the issuance of each security is constant over time and second that since  $p_H^{N-1} < p_L^{N-1}$  then  $B^1 < 0$  and  $B^N > 0$ .

## A.2 Coupon Bonds and No Buyback: the Ramsey Program

We solve the optimal policy problem under no buyback and coupons assuming that the government issues debt in one year and  $N$  year bonds. As in the rest of the paper we introduce debt limits, these are parameterized as:

$$(8) \quad b_t^N \in \left[ \frac{\underline{M}_N}{\sum_{j=1}^N \beta^j + \kappa \sum_{j=1}^N \sum_{i=1}^j \beta^i}, \frac{\overline{M}_N}{\sum_{j=1}^N \beta^j + \kappa \sum_{j=1}^N \sum_{k=1}^j \beta^k} \right] \equiv [\underline{\widetilde{M}}_N, \widetilde{\overline{M}}_N]$$

Letting  $[\underline{\widetilde{M}}_1, \widetilde{\overline{M}}_1] \equiv [\frac{\underline{M}_1}{\beta}, \frac{\overline{M}_1}{\beta}]$  be the analogous constraint set for one year debt the planning problem is given by:

$$\begin{aligned} \mathcal{L} = E_0 \sum \beta^t & \left\{ u(c_t) + v(T - c_t - g_t) + \lambda_t \left[ b_t^1 \beta u_{c,t+1} + b_t^N \left( \beta^N u_{c,t+N} + \sum_{j=1}^N \beta^j u_{c,t+j} \kappa \right) \right. \right. \\ & \left. \left. - b_{t-1}^1 u_{c,t} - b_{t-N}^N u_{c,t} - \kappa \sum_{j=1}^N b_{t-j}^i u_{c,t} - g_t u_{c,t} + (u_{c,t} - v_{x,t})(g_t + c_t) \right] \right. \\ & \left. + \sum_{i \in \{1, N\}} \xi_{U,t}^i (\widetilde{M}_i - b_t^i) + \sum_{i \in \{1, N\}} \xi_{L,t}^i (b_t^i - \widetilde{M}_i) \right\} \end{aligned}$$

The first order condition for consumption is :

$$\begin{aligned} & u_{c,t} - v_{x,t} + \lambda_t (u_{cc,t} c_t + u_{c,t} + v_{xx,t} (c_t + g_t) - v_{x,t} \\ & + u_{cc,t} \kappa \sum_{j=1}^N (\lambda_{t-j} - \lambda_t) b_{t-j}^N + u_{cc,t} \sum_{i \in \{1, N\}} (\lambda_{t-i} - \lambda_t) b_{t-i}^i = 0 \end{aligned}$$

and off corners the analogous conditions for  $b_t^1$  and  $b_t^N$  are:

$$(9) \quad \lambda_t E_t(u_{c,t+1}) = E_t(\lambda_{t+1} u_{c,t+1})$$

$$(10) \quad \lambda_t E_t(\kappa \sum_{j=1}^N \beta^j u_{c,t+j} + \beta^N u_{c,t+N}) = E_t(\kappa \sum_{j=1}^N \beta^j u_{c,t+j} \lambda_{t+j} + \beta^N u_{c,t+N} \lambda_{t+N})$$

For brevity we summarized the properties of this model in Table 4 in text. In Figure 1 of this appendix we show a typical sample of long and short debt (analogous to Figures 5-9 in the main text). As was claimed in text, when bonds pay positive coupons the properties of the model remain very close to the no buyback and zero coupons case.

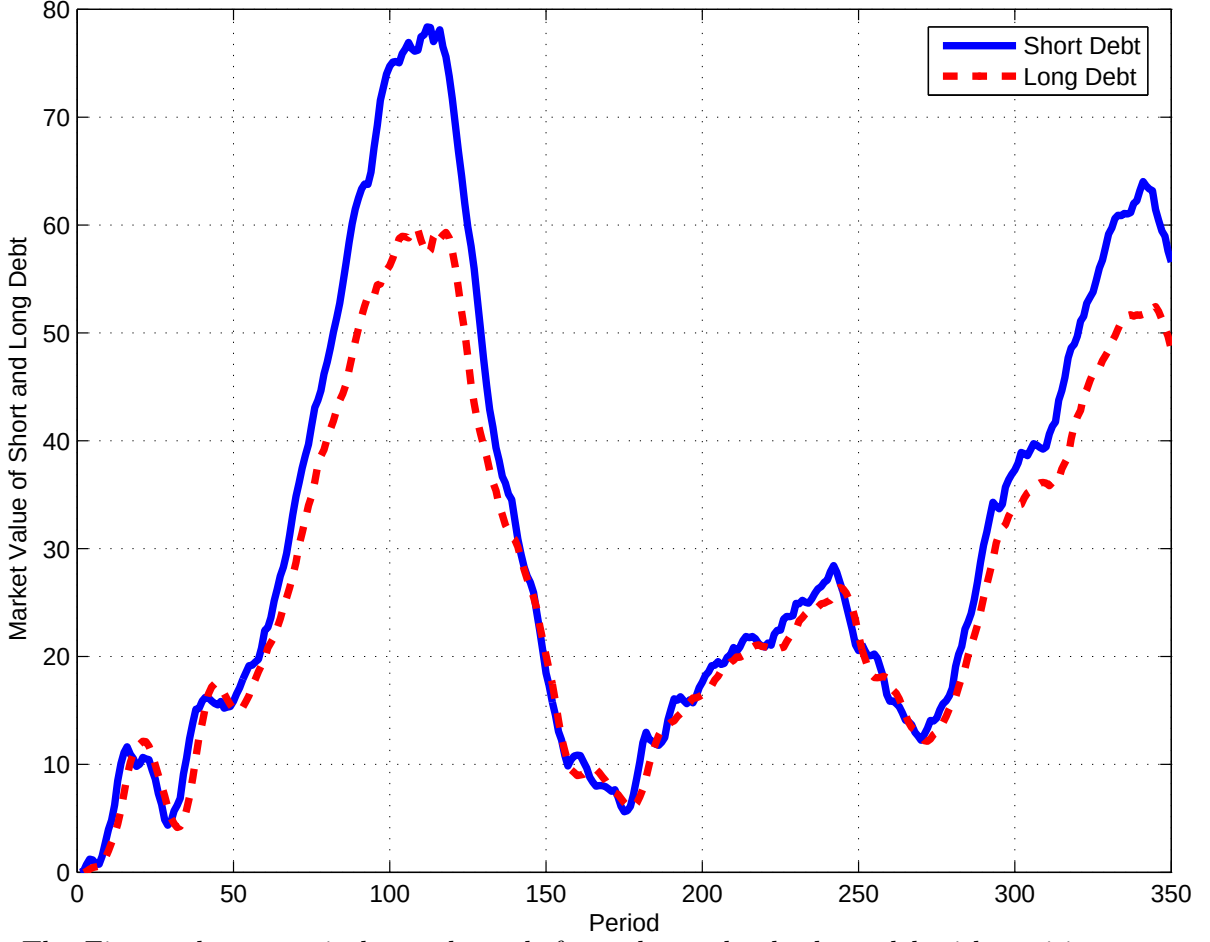
### A.3 Callable Bonds: the Ramsey Program

As explained in Section 2 of the paper the US government issued callable bonds in the past. These types of securities give to the issuer the option to buy them back after  $m$  years, at every coupon date until the bond matures. Their price at buyback is at par. We showed that historically the US government has repurchased callable bonds at the start of the call window.

Proposing this model, our intention is not to motivate the empirical observations of why the Treasury chooses to buyback at the first call date. Rather we seek to establish that removing debt from the market before, but close to, the maturity date is akin to the model of no buyback and that our findings about the importance of short term debt and the positive comovement between short and long debt still hold.

We solved our model assuming that the buyback of the  $N$ -year bond occurs  $m$  years after issuance. To make our analysis as conservative as possible we calibrated  $N = 10$  and  $m = 5$ . Notice that a lower  $m$  makes the model closer to the buyback section, since buyback is equivalent with  $m = 1$ . If we were to find that even for a low  $m$  the model behaves similar to no buyback, higher  $m$  are likely to be even closer to no-buyback. The call window in the data for 10 year bonds starts 2 years before maturity, suggesting a buyback period of  $m = 8$ . In this sense, our model choice of  $m = 5$  is

Figure 1: Market Value of Short and Long Debt under no Buyback+Coupons



Notes: The Figure plots a typical sample path from the no buyback model with positive coupons. As explained in text the value of the coupon  $\kappa$  is calibrated so that bonds trade on average at par. The upper bound on  $b_t^N$  is given in (8). The lower bound is zero. The value of short term debt in a given period  $t$  in the Figure, is constructed by adding the coupon payments and principals which are to mature in  $t + 1$  to the market value of one year bonds issued in  $t$ .

conservative.

The budget constraint of the government is:

$$\sum_{i \in \{1, N\}} p_t^i b_t^i = b_{t-1}^1 + p_t^{N-m} b_{t-m}^N + g_t - \tau_t(T - x_t)$$

The ad hoc debt constraints for the  $N$  year bond are

$$b_t^N \in \left[ \frac{\underline{M}_N}{\sum_{j=0}^{m-1} \beta^{N-j}}, \frac{\overline{M}_N}{\sum_{j=0}^{m-1} \beta^{N-j}} \right] \equiv [\widetilde{\underline{M}}_N, \widetilde{\overline{M}}_N]$$

Letting  $[\widetilde{\underline{M}}_1, \widetilde{\overline{M}}_1] = [\frac{\underline{M}_1}{\beta}, \frac{\overline{M}_1}{\beta}]$  be the analogous constraints for one year debt and substituting the equilibrium expressions for the tax rate and the bond prices we represent the planning problem as

follows:

$$\begin{aligned} \mathcal{L} = E_0 \sum_t \beta^t & \left\{ u(c_t) + v(T - c_t - g_t) + \lambda_t \left[ \sum_{i \in \{1, N\}} b_t^i \beta^i u_{c,t+i} - b_{t-1}^1 u_{c,t} - b_{t-m}^N \beta^{N-m} u_{c,t+N-m} \right. \right. \\ & \left. \left. - g_t u_{c,t} + (u_{c,t} - v_{x,t})(g_t + c_t) \right] \right. \\ & \left. + \sum_{i \in \{1, N\}} \xi_{U,t}^i (\widetilde{M}_i - b_t^i) + \sum_{i \in \{1, N\}} \xi_{L,t}^i (b_t^i - \widetilde{M}_i) \right\} \end{aligned}$$

The first order conditions for the optimum are given by:

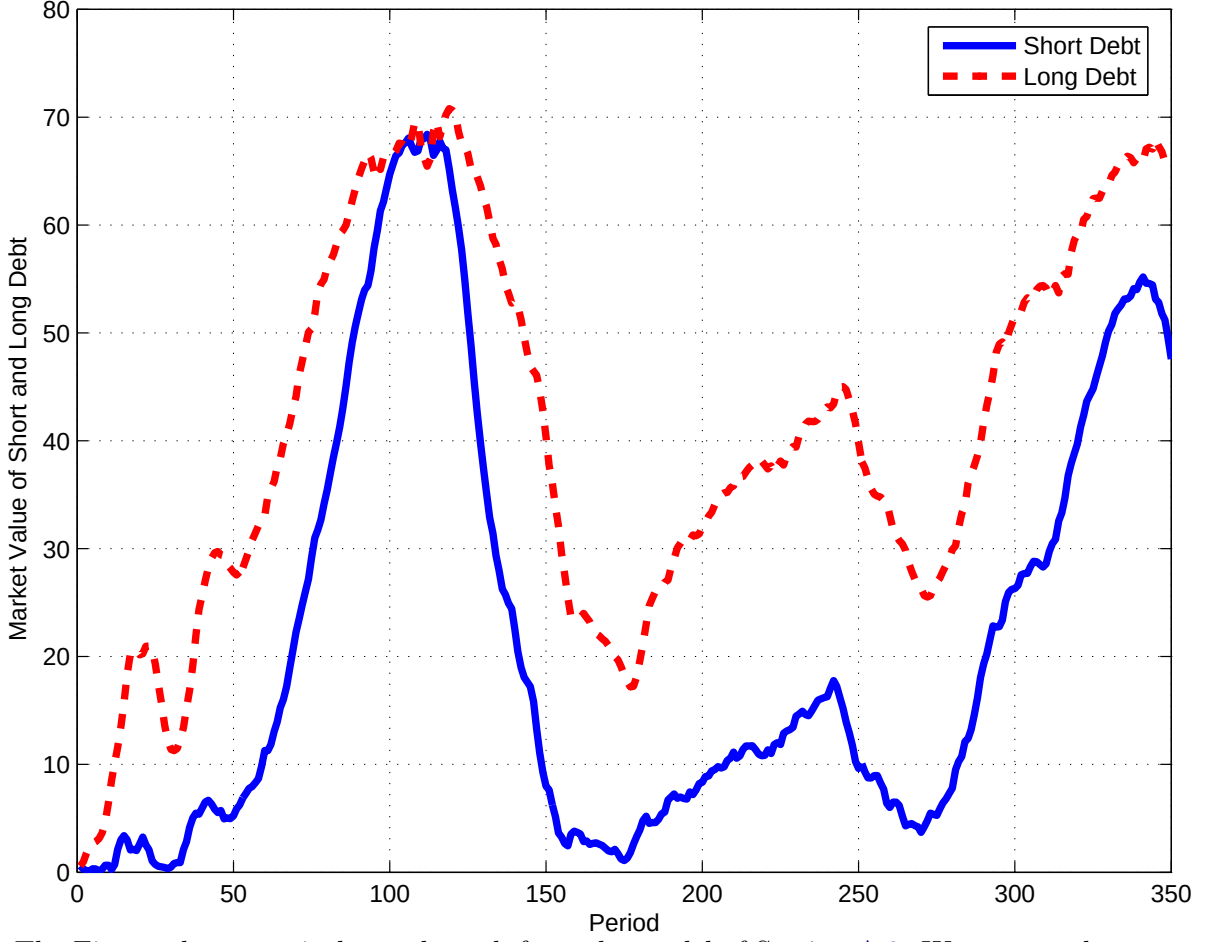
$$\begin{aligned} u_{c,t} - v_{x,t} + \lambda_t & \left( u_{cc,t} c_t + u_{c,t} + v_{xx,t}(c_t + g_t) - v_{x,t} \right) + u_{cc,t} \left[ (\lambda_{t-1} - \lambda_t) b_{t-1}^1 + (\lambda_{t-N} - \lambda_{t-N+m}) b_{t-N}^N \right] \\ & \beta E_t(u_{c,t+1} \lambda_t - u_{c,t+1} \lambda_{t+1}) + \xi_{L,t}^1 - \xi_{U,t}^1 = 0 \\ & \beta^N E_t(u_{c,t+N} \lambda_t - u_{c,t+N} \lambda_{t+m}) + \xi_{L,t}^N - \xi_{U,t}^N = 0 \end{aligned}$$

We assume  $\widetilde{M}_1 = \widetilde{M}_N = 0$ . In Figure 2 we plot a typical sample of the market value of short and long debt. Notice that indeed assuming that the government repurchases debt from the market  $m = 5$  years after issuance, reduces the share of short bonds in the portfolio compared to a model with no buyback. The portfolio is somewhere between the ‘full buyback model’ studied in the paper (i.e. when  $m = 1$ ) and the no buyback model where  $m = N$ . In terms of the moments reported in Table 4 in the paper the model of this section gives us the following:  $\overline{\mathcal{S}}_t = 19\%$ ,  $\sigma_{\mathcal{S}_t} = 11.6\%$ ,  $\rho_{\mathcal{S}_t, \mathcal{S}_{t-1}} = 0.84$   $\rho_{\widetilde{b}_t^S, \widetilde{b}_t^N} = 0.62$   $\%_{\mathcal{S}_t=0} = 0.65\%$ .

We view these results as encouraging because they confirm the hypothesis that even if the government buys back som of the debt before maturity there is still a role for short bonds. First, because the share of short debt is very rarely zero in simulations (e.g.  $\%_{\mathcal{S}_t=0} = 0.65\%$  versus the analogous figure in the buyback no lending model in the paper of 13%). As we mentioned, the choice of  $m = 5$  is quite conservative. The data suggest that  $m = 8$  would be more appropriate. With  $m = 8$  we expect the model to generate results very close to the no buyback ones.

This is only a partial study of callable bonds. Clearly, the modelling of callable bonds can be made closer to the data by introducing that they can be repurchased at par or the transaction costs that are involved in the recall. A model taking all these features into account is beyond the scope of this paper. However the message that there is still a role for short bonds comes out clearly from the analysis.

Figure 2: Market Value of Short and Long Debt under 'Callable' Bonds



Notes: The Figure plots a typical sample path from the model of Section A.3. We assume that government buybacks of 10 year bonds occur 5 years after issuance.

#### A.4 Optimal Repurchases: the Ramsey Program

In the optimal repurchase (OR) model of Section 6 the government maximizes the utility of the household subject to the following constraints

$$(11) \quad \sum_{i \in \{S, N\}} p_t^i b_t^i (1 - \mathcal{T}^i(b_t^i)) = b_{t-S}^S + b_{t-N}^N - R_{t-N+1} + p_t^{N-1} R_t (1 + \mathcal{T}^R(R_t)) + g_t - \tau_t(T - x_t)$$

$$(12) \quad T - x_t = c_t + g_t + \mathcal{TC}_t$$

$$(13) \quad 0 \leq b_t^i \leq \frac{\bar{M}_i}{\sum_{j=1}^i \beta^j}, \quad 0 \leq R_t \leq b_{t-1}^N$$

where  $\mathcal{TC}_t \equiv \sum_{i \in \{S, N\}} p_t^i b_t^i \mathcal{T}^i(b_t^i) + p_t^{N-1} R_t (1 + \mathcal{T}^R(R_t))$ .

To simplify the solution of this model we assume that the government treats as exogenous the function  $\mathcal{TC}_t$ , in other words it does not take derivatives of  $\mathcal{TC}_t$  with respect to consumption and

the bonds.<sup>1</sup>

The Lagrangian of the OR model is:

$$(14) \quad \mathcal{L} = E_0 \sum_t \beta^t \left[ u(c_t) + v(T - c_t - g_t - \mathcal{T}C_t) + \lambda_t \left[ \sum_{i \in \{S, N\}} b_t^i \beta^i u_{c,t+i} (1 - \mathcal{T}(b_t^i)^i) \right. \right. \\ \left. \left. - \beta^{N-1} u_{c,t+N-1} R_t (1 + \mathcal{T}^R(R_t)) - (b_{t-S}^S + b_{t-N}^N - R_{t-N+1}) u_{c,t} \right. \right. \\ \left. \left. - g_t u_{c,t} + (u_{c,t} - v_{x,t})(g_t + c_t) \right] + \sum_{i \in \{S, N\}} \xi_{U,t}^i \left( \frac{\bar{M}_i}{\sum_{j=1}^i \beta^j} - b_t^i \right) + \sum_{i \in \{S, N\}} \xi_{L,t}^i b_t^i + \xi_{U,t}^R (b_{t-1}^N - R_t) + \xi_{L,t}^R R_t \right]$$

The FONC are given by:

$$\begin{aligned} u_{c,t} - v_{x,t} + \lambda_t (-u_{cc,t} g_t + u_{c,t} + u_{cc,t} (T - x_t) + v_{xx,t} (T - x_t) - v_{x,t}) - u_{cc,t} [B_{t-S} \lambda_t - B \lambda_{t-S}] &= 0 \\ E_t \beta (-u_{c,t+1} \lambda_{t+1} + u_{c,t+1} \lambda_t (1 - \mathcal{T}_t^1 - \mathcal{T}_{b_t^1}^1 b_t^1)) + \xi_{L,t}^1 - \xi_{U,t}^1 &= 0 \quad \text{for } i = S \\ E_t \beta^N (-u_{c,t+N} \lambda_{t+N} + u_{c,t+N} \lambda_t (1 - \mathcal{T}_t^N - \mathcal{T}_{b_t^N}^N b_t^N)) + E_t \beta (\xi_{U,t+1}^R) + \xi_{L,t}^N - \xi_{U,t}^N &= 0 \quad \text{for } i = N \\ E_t \beta^{N-1} (u_{c,t+N-1} \lambda_{t+N-1} - u_{c,t+N-1} \lambda_t (1 + \mathcal{T}_t^R + \mathcal{T}_{R_t}^R R_t)) + \xi_{L,t}^R - \xi_{U,t}^R &= 0 \end{aligned}$$

where

$$\begin{aligned} B_t &\equiv b_t^S + B_{t-N+1+S}^{net} \\ B_t^{net} &\equiv b_{t-1}^N - R_t \\ B \lambda_t &\equiv \lambda_t (1 - \mathcal{T}_t^S) b_t^S + B \lambda_{t-N+1+S}^{net} \\ B \lambda_t^{net} &\equiv \lambda_{t-1} (1 - \mathcal{T}_{t-1}^N) b_{t-1}^N - \lambda_t (1 + \mathcal{T}_t^R) R_t \end{aligned}$$

## B Numerical Appendix

In section 4 of the paper we described the ‘Condensed PEA’ that deals with the high dimensionality of the state vector, and the ‘Forward States PEA’ that deals with the indeterminacy of the portfolio generated by the use of the PEA. This numerical appendix outlines in greater detail the two methods, their implementation and the steps we followed to approximate the conditional expectations in the different models. In particular, we report how we selected the state variables of the core vector,  $X_t^{core}$ , the ‘out’-vector,  $X_t^{out}$  and the order of the polynomials of the states that were used. Moreover, we report how many linear combinations of state variables were added to the approximations, and also discuss some practical features of our numerical procedure that can help the algorithm convergence.

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<sup>1</sup>Without this assumption we would need to keep track of the fact that there is a conditional expectation in the determination of  $\mathcal{T}C_t$ , therefore the solution to the model would feature both current and lagged values of the multiplier on this constraint. This would add yet more state variables in the model but with minimal quantitative effects.

Note that another way to simplify the planner’s program (avoid having to keep track of the resource constraint as a separate object in the Lagrangean) is to assume  $\mathcal{T}C_t$  do not enter the feasibility constraint. In this case transaction costs do not impact the overall resources of the economy, this would correspond to a situation where a financial firm can charge transaction costs on bond issuances without actually spending labor resources in it. When we run the model under this assumption (which appears sometimes in the literature) we found virtually no effect on our results.



## B.1 Implementation of "Condensed PEA" and "Forward states PEA"

### B.1.1 Selection of variables in the approximation

Recall that 'Condensed PEA' divides the state vector  $X$  in two subvectors: the core vector,  $X_t^{core}$ , which includes variables that (we believe a priori) are of primary importance in the approximation, and the  $X_t^{out}$  vector, which includes the remaining state variables and possibly higher order polynomial terms. 'Forward States PEA' resolved the portfolio indeterminacy issue through approximating  $E_t u_{c,t+i}$  with  $E_t(\Phi^i(X_{t+1}, \gamma^i))$  and  $E_t \lambda_{t+1} u_{c,t+i}$  with  $E_t(\Psi^i(X_{t+1}, \delta^i))$ . To clearly show how we implemented these two methods, we bring them now together and in what follows we outline the 'Condensed PEA' using since the first PEA iteration (i.e. with core state variables only) 'Forward States' to solve the portfolio choice.

#### The $X^{core}$ vector:

In all the models presented in the paper, but the 3 bond model, we used the core vector

$$(15) \quad X_{t+1}^{core} = \left\{ 1, g_{t+1}, \{b_t^i\}_{i=1,N}, \lambda_t, \{(b_t^i)^2\}_{i=1,N}, \{g_{t+1} b_t^i\}_{i=1,N} \right\}$$

i.e.  $X_{t+1}^{core}$  is composed of a constant, the level of government spending in  $t+1$ , the levels of date  $t$  variables (the bonds and the multipliers), the square of the bonds and the interaction term between the bonds in  $t$  and  $g_{t+1}$ .

We solve the system of FONCs after integrating out the term  $g_{t+1}$  as discussed in footnote 25 of the main text. We use the analytical formula for the conditional expectation of  $g_{t+1}$  at time  $t$  given by:

$$(16) \quad \int g_{t+1} f_{g_{t+1}|g_t} dg_{t+1} = \omega_t + (\underline{g} - \omega_t) \Phi\left(\frac{\underline{g} - \omega_t}{\sigma_\epsilon}\right) + (\bar{g} - \omega_t) \left[1 - \Phi\left(\frac{\bar{g} - \omega_t}{\sigma_\epsilon}\right)\right] - \sigma_\epsilon \left[ \phi\left(\frac{\bar{g} - \omega_t}{\sigma_\epsilon}\right) - \phi\left(\frac{\underline{g} - \omega_t}{\sigma_\epsilon}\right) \right]$$

where  $\omega_t \equiv \rho_g g_t + (1 - \rho_g) g_{ss}$ ,  $\Phi(\phi)$  is the standard normal cdf (pdf),  $\bar{g}$  and  $\underline{g}$  are upper and lower bounds on government spending<sup>2</sup>.  $\sigma_\epsilon$  is the standard deviation of the spending shock.<sup>3</sup>

We have chosen to introduce higher order terms in the core vector for three reasons. First, approximating the conditional expectations  $E_t u_{c,t+i}$  means that we are approximating the bond

<sup>2</sup>As discussed in the calibration section we assume that spending fluctuates between 15 and 35 percent of steady state GDP.

<sup>3</sup>The expression is reached as follows:

$$\begin{aligned} \int g_{t+1} f_{g_{t+1}|g_t} dg_{t+1} &= \int_{-\infty}^{\underline{g}-\omega_t} \underline{g} dF(\epsilon_{t+1}) + \int_{\bar{g}-\omega_t}^{\infty} \bar{g} dF(\epsilon_{t+1}) + \int_{\underline{g}-\omega_t}^{\bar{g}-\omega_t} (\omega_t + \epsilon_{t+1}) dF(\epsilon_{t+1}) \\ &= \omega_t + \int_{-\infty}^{\underline{g}-\omega_t} (\underline{g} - \omega_t) dF(\epsilon_{t+1}) + \int_{\bar{g}-\omega_t}^{\infty} (\bar{g} - \omega_t) dF(\epsilon_{t+1}) + \int_{\underline{g}-\omega_t}^{\bar{g}-\omega_t} \epsilon_{t+1} dF(\epsilon_{t+1}) \end{aligned}$$

where  $F$  denotes the cdf of  $\epsilon$ . Standard results give:

$$\int_{-\infty}^{\underline{g}-\omega_t} (\underline{g} - \omega_t) dF(\epsilon_{t+1}) = \int_{-\infty}^{\frac{\underline{g}-\omega_t}{\sigma_\epsilon}} (\underline{g} - \omega_t) \frac{1}{\sqrt{2\pi}} e^{-\frac{z_{t+1}^2}{2}} dz_{t+1} = (\underline{g} - \omega_t) \Phi\left(\frac{\underline{g} - \omega_t}{\sigma_\epsilon}\right)$$

where  $z$  is a standard normal variable and  $\Phi$  is the cdf. Analogously:

$$\int_{\bar{g}-\omega_t}^{\infty} (\bar{g} - \omega_t) dF(\epsilon_{t+1}) = (\bar{g} - \omega_t) (1 - \Phi\left(\frac{\bar{g} - \omega_t}{\sigma_\epsilon}\right))$$

prices. If we include only the levels of the bonds,  $\{b_t^i\}_{i=1,N}$ , we are imposing that bonds are close substitutes in terms of their influence on prices.<sup>4</sup> This is not a property that we are sure to find in equilibrium and non-linear terms may be potentially important. Secondly, if the ad hoc debt limits are occasionally binding this is known to introduce non-linearities so that higher order terms may be important. Finally the ‘Condensed PEA’ inclusion criterion suggested that these non linear terms are important for the approximations. Indeed when we solved the models without higher order terms and tested whether they should be included in the linear combinations, the percentage gains in  $R^2$  were substantial. Higher order terms therefore should be included in the polynomials, either in  $X_{t+1}^{core}$  or in  $X_{t+1}^{out}$ . We ultimately chose to introduce some higher order terms in the core vector and left others for the linear combinations (see below) finding this helpful for the stability of the convergence.

### The $\mathbf{X}^{out}$ vector:

Notice that if we think of introducing higher-order terms  $\mathbf{X}^{out}$  is potentially infinite-dimensional, so we need to specify what we actually included in  $\mathbf{X}^{out}$  in our simulations, as the linear combinations of this vector are the ones that eventually can play a role in the solution.

Assuming  $S = 1$ , the ‘out’ vector is different for each model presented in the paper. To identify the elements of  $X^{out}$  we use as guidance the FONC of each model.

Consider first the *buyback model*. The first order conditions are

$$(17) \quad u_{c,t} - v_{x,t} + \lambda_t \left( u_{cc,t} c_t + u_{c,t} + v_{xx,t} (c_t + g_t) - v_{x,t} \right) + u_{cc,t} \sum_{i \in \{1,N\}} \left( \lambda_{t-i} - \lambda_{t-i+1} \right) b_{t-i}^i = 0$$

$$(18) \quad \beta^i E_t \left( u_{c,t+i} \lambda_t - u_{c,t+i} \lambda_{t+1} \right) + \xi_{L,t}^i - \xi_{U,t}^i = 0 \quad \text{for } i = 1, N$$

When markets are incomplete, the term  $\sum_{i \in \{1,N\}} \left( \lambda_{t-i} - \lambda_{t-i+1} \right) b_{t-i}^i$  summarises interest rate manipulation under commitment (see FMOS (2016)). Suppose that a positive spending shock arrives in period  $t$  and that  $b_t^N > 0$ . Since  $\left( \lambda_{t-1} - \lambda_t \right) b_t^N$  becomes negative, the government finds optimal to promise a tax cut in  $t + N - 1$  and lower the marginal utility of consumption in that period. It is then evident that the terms  $\lambda_{t-1} b_t^N$  and  $\lambda_t b_t^N$  are important determinants of  $u_{c,t+N-1}$  and  $u_{c,t+N}$  and hence they should be accounted for when we approximate the conditional expectations.<sup>5</sup>

Finally,

$$\int_{\underline{g}-\omega_t}^{\bar{g}-\omega_t} \epsilon_{t+1} dF(\epsilon_{t+1}) = \int_{\frac{\underline{g}-\omega_t}{\sigma_\epsilon}}^{\frac{\bar{g}-\omega_t}{\sigma_\epsilon}} \sigma_\epsilon \frac{1}{\sqrt{2\pi}} z_{t+1} e^{-\frac{z_{t+1}^2}{2}} d\epsilon_{t+1} = -\sigma(\phi(\frac{\bar{g}-\omega_t}{\sigma_\epsilon}) - \phi(\frac{\underline{g}-\omega_t}{\sigma_\epsilon}))$$

Putting everything together we get (16)

<sup>4</sup>To see this consider the approximation of  $E_t u_{c,t+N} \approx \gamma_0^N + \gamma_1^N \int g_{t+1} f(g_{t+1}|g_t) dg_{t+1} + \gamma_2^N b_t^1 + \gamma_3^N b_t^N + \gamma_4^N \lambda_t$  under linear polynomials. Clearly there are (infinitely) many pairs  $(b_t^1, b_t^N)$  that give the same bond price (holding  $\lambda_t$  fixed).

Notice that the optimal portfolio is nonetheless identified under linear polynomials since  $\{b_t^i\}_{i=1,N}$  influence all conditional expectations and enter in a nonlinear fashion in the system of FONCs (for example in the budget constraint of the government).

<sup>5</sup>In the text the implementation of ‘Forward States’ to the buyback model was summarized in the following equations

$$(19) \quad \lambda_t = \frac{E_t(\Psi^i(X_{t+1}, \delta^i))}{E_t(\Phi^i(X_{t+1}, \gamma^i))} \quad \text{for } i = S, N$$

$$(20) \quad \sum_{i \in \{S,N\}} b_t^i \beta^i E_t(\Phi^i(X_{t+1}, \gamma^i)) = \sum_{i \in \{S,N\}} b_{t-1}^i \beta^{i-1} \Phi^i(X_t, \gamma^i) + g_t u_{c,t} - (u_{c,t} - v_{x,t})(g_t + c_t)$$

Applying the above argument to determine which states potentially exert a significant influence to the expectations of date  $t + 1$ ,  $t + N - 1$  and  $t + N$  variables in the buyback model, we include in  $X_{t+1}^{out}$  the following terms:  $\lambda_t b_t^N$ ,  $\lambda_t b_t^1$ ,  $\lambda_t b_{t-1}^N$ ,  $\lambda_{t-1} b_{t-1}^N$ ,  $\lambda_{t-N+1} b_{t-N+1}^N$  and  $\lambda_{t-N+2} b_{t-N+1}^N$ .

Two more comments about this choice are necessary. Firstly, despite the fact that each of the above terms is potentially important for (some of) the conditional expectations we wish to approximate, it is unlikely that each term bears the same importance to each conditional expectation. For example, the term  $\lambda_t b_t^N$  clearly exerts an influence on  $u_{c,t+N}$  (through the FONC) but it is less likely to exert a significant influence on  $u_{c,t+N-1}$ . In this case the ‘Condensed PEA’ will assign a coefficient close to zero to  $\lambda_t b_t^N$  in the approximation of  $E_t u_{c,t+N-1}$  and a coefficient different from zero in the approximation of  $E_t u_{c,t+N}$ . This shows how convenient it is to include these terms in  $X^{out}$  where having coefficients close to zero for some state variables is not an issue, as opposed to including them in  $X^{core}$ , in which case variables with close to zero coefficients may cause convergence problems.

Secondly, as explained before, (17) suggests that the cross terms between  $\lambda$  and  $b$  are potentially important for the solution. However, one may wonder whether the levels of these variables should also be included in the state vector. The FONCs show that the influence of  $\lambda_{t-N+1}$  on the optimal allocation in  $t + 1$  is close to zero if  $b_{t-N+1}^N$  is close to zero. The effect of changes in the value of the multiplier is felt more when government debt is high. This nonlinear influence seems to be (sufficiently) well captured in our specification by the cross terms and not by the levels since, as we verify in section 7.4 of the main text, we pass accuracy tests.<sup>6</sup>

We apply the above selection criterion to the other models. Consider the *no buyback model* and its first order conditions and budget constraint:

$$\begin{aligned} u_{c,t} - v_{x,t} + \lambda_t \left( u_{cc,t} c_t + u_{c,t} + v_{xx,t} (c_t + g_t) - v_{x,t} \right) + u_{cc,t} \sum_{i \in \{1,N\}} \left( \lambda_{t-i} - \lambda_t \right) b_{t-i}^i &= 0 \\ \beta^i E_t \left( u_{c,t+i} \lambda_t - u_{c,t+i} \lambda_{t+i} \right) + \xi_{L,t}^i - \xi_{U,t}^i &= 0 \quad \text{for } i = 1, N \\ \sum_{i \in \{1,N\}} b_t^i \beta^i E_t u_{c,t+i} &= g_t u_{c,t} + u_{c,t} \sum_{i=1,N} b_{t-i}^i - (u_{c,t} - v_{x,t}) (c_t + g_t). \end{aligned}$$

We include in the  $X^{out}$  vector:  $\lambda_t b_t^N$ ,  $\lambda_t b_t^1$ ,  $\lambda_{t-N+1} b_{t-N+1}^N$ , and  $b_{t-N+1}^N$ , as these appear directly on the FONC.

Next, consider the *no buyback model with coupons*. To solve the coupon model we need to approximate the term  $\sum_{j=1}^N \beta^j E_t u_{c,t+j} \kappa + \beta^N u_{c,t+N}$  and the term  $\sum_{j=1}^N \beta^j E_t u_{c,t+j} \lambda_{t+j} \kappa + \beta^N u_{c,t+N} \lambda_{t+N}$ .

Notice that in (20) we parameterize the term  $E_t u_{c,t+N-1}$  as  $\Phi^N(X_t, \gamma^N)$ . In other words we apply the standard PEA to this term. An alternative is to define  $E_t u_{c,t+N-2} = \Phi^N(X_t, \gamma^{N-1})$  and then use Forward States to get:  $E_t u_{c,t+N-1} = E_t \Phi^N(X_{t+1}, \gamma^{N-1})$ . We follow the latter route in the numerical implementation. We therefore write (20) as follows

$$(21) \quad b_t^1 \beta E_t (\Phi^1(X_{t+1}, \gamma^1)) + b_t^N \beta^N E_t (\Phi^N(X_{t+1}, \gamma^N)) = b_{t-1}^1 u_{c,t} + b_{t-1}^N \beta^{N-1} E_t (\Phi^{N-1}(X_{t+1}, \gamma^{N-1})) + g_t u_{c,t} - (u_{c,t} - v_{x,t}) (g_t + c_t)$$

i.e. when  $S = 1$  and realizing that  $E_t u_{c,t} = u_{c,t} = \Phi^1(X_t, \gamma^1)$ .

The two ways of solving the model are obviously conceptually equivalent.

<sup>6</sup>Recall that  $X^{core}$  includes the variables  $\lambda_t, b_t^N$  and  $b_{1,t}$  in levels. These first order terms, help us to identify the portfolio, but combined with their squares, cubes and so on can (practically speaking), explain part of the variability of some of the cross terms in  $X^{out}$ . To avoid having residuals close to zero from the regressions of  $X^{out}$  on  $X^{core}$  when we compute linear combinations, we use an additional selection criterion that we describe in the next subsection.

From the FONC of consumption and the government budget constraint (omitted for brevity), it is easy to show that all the lags of  $b_{t-j}^N$  and  $\lambda_{t-j}b_{t-j}^N$ , for  $j = 1, 2, \dots, N-1$  should be introduced in the out vector. The  $X^{out}$  vector is therefore composed by:  $\{\lambda_{t-i}b_{t-i}^N\}_{i=0}^{N-1}$ ,  $\lambda_t b_t^1$ ,  $\{b_{t-i}^N\}_{i=1}^{N-1}$ .

Similarly, when we consider the *callable bond* model using the same criteria discussed above we know that the repurchase date,  $m$ , will exert an influence on the solution and then we have chosen the  $X^{out}$  vector to be:  $\{b_t^i \lambda_t\}_{i=1, N}$ ,  $\lambda_t b_{t-N+m}^N$ ,  $\lambda_{t-N+1} b_{t-N+1}^N$ ,  $\lambda_{t-N+m} b_{t-N+1}^N$  and  $b_{t-N+m+1}^N$ .

Moreover for each model discussed we include in  $X^{out}$  other higher order terms of date  $t$  variables that have not been included in  $X^{core}$ . In each approximation we add in  $X^{out}$  the following terms:  $\lambda_t^2$ ,  $(b_t^N)^3$ ,  $(b_t^1)^3$ ,  $b_t^N b_t^1$ .

We now consider the *optimal repurchases* of section 6.3 in the main text, which FONCs shown in a previous subsection. The following expectations need to be approximated with PEA in this case:

$$E_t \xi_{U,t+1}^R \quad \text{and} \quad E_t u_{c,t+i}, \quad E_t \lambda_{t+i} u_{c,t+i}, \quad i = 1, N, N-1$$

where  $\xi_{U,t}^R$  is the Langrange multiplier on the constraint  $R_t \leq b_{t-1}^N$ .

As discussed in the text, one way to rewrite the state vector and reduce the total number of state variables is to redefine the state vector as:

$$(22) \quad X_{t+1} = \left[ g_{t+1}, B_t, B\lambda_t, (B_{t+1-i}^{net}, B\lambda_{t+1-i}^{net})_{i=1}^N, \lambda_{t-N}, b_{t-N}^N \right]$$

where

$$\begin{aligned} B_t^{net} &\equiv b_{t-1}^N - R_t \\ B_t &\equiv b_t^S + B_{t-N+2}^{net} \\ B\lambda_t^{net} &\equiv \lambda_{t-1}(1 - \mathcal{T}^N)b_{t-1}^N + \lambda_t(1 + \mathcal{T}^R)R_t \\ B\lambda_t &\equiv \lambda_t(1 - \mathcal{T}^1)b_t^1 + B\lambda_{t-N+2}^{net} \end{aligned}$$

As for the other models we chose  $X^{core}$  and  $X^{out}$  to include the state variables which appear in the FONC of the model and which therefore exert a direct influence on the conditional expectations. We specified  $X^{core}$  as in (15) and  $X^{out}$  as follows:

$$(23) \quad X_{t+1}^{out} = \left\{ \{b_t^i \lambda_t\}_{i=1, N}, b_{t-N+1}^N - R_{t-N+2}, (b_{t-N+1}^N - R_{t-N+2})\lambda_{t-N+1}, R_{t-N+2}\lambda_{t-N+2} \right\}.$$

It is worth noting that  $B_t, B\lambda_t, (B_{t+1-i}^{net}, B\lambda_{t+1-i}^{net})_{i=1}^N$  are combinations of the variables included in this specification of  $X_{t+1}^{out}$ . For example  $B_t = b_t^1 + b_{t-N+1}^N - R_{t-N+2}$  and  $B_{t-N+2}^{net} = b_{t-N+1}^N - R_{t-N+2}$  are part of the state vector. Therefore we are using the state variables that are sufficient. We have chosen to separate the terms  $b_t^1$  and  $b_{t-N+1}^N - R_{t-N+2}$  in the approximations assigning  $b_t^1$  to the core and  $b_{t-N+1}^N - R_{t-N+2}$  to the out vector. We did this for convenience and most importantly to be able to use as initial guess for our approximation the solution of the no buyback model.

Moreover, notice that though in principle we could introduce  $R_t$  as a variable in  $X^{core}$ ,<sup>7</sup> this is not necessary to identify the optimal path of  $R_t$ . Since this is a model where the government can

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<sup>7</sup>From (22) we know that  $b_{t-1}^N - R_t$  is a state variable. However, this will not appear in the FONC in periods  $t+1, t+N, t+N-1$  and for this reason we dropped it from the core state vector and from the out vector (23).

repurchase only after one period and we assume positive transaction costs, we do not need  $R_t$  in the core states to determine the portfolio.<sup>8,9</sup>

Finally notice that in  $X^{core}$  and  $X^{out}$ , the bond and repurchases variables are not multiplied by transaction costs. Since these variables (mostly) enter separately in the approximations and since the costs  $\mathcal{T}$  are small, this does not influence the properties of the solution.

Let's now turn to the *model with three bonds*. When the government issues debt in three maturities ( $1 < M < N$ ) under no buyback the FONC are given by:

$$\begin{aligned} u_{c,t} - v_{x,t} + \lambda_t \left( u_{cc,t} c_t + u_{c,t} + v_{xx,t} (c_t + g_t) - v_{x,t} \right) + u_{cc,t} \sum_{i \in \{1, M, N\}} \left( \lambda_{t-i} - \lambda_t \right) b_{t-i}^i &= 0 \\ \beta^i E_t \left( u_{c,t+i} \lambda_t - u_{c,t+i} \lambda_{t+i} \right) + \xi_{L,t}^i - \xi_{U,t}^i &= 0 \quad \text{for } i = 1, M, N \\ \sum_{i \in \{1, M, N\}} b_t^i \beta^i E_t u_{c,t+i} &= g_t u_{c,t} + u_{c,t} \sum_{i \in \{1, M, N\}} b_{t-i}^i - (u_{c,t} - v_{x,t}) (c_t + g_t). \end{aligned}$$

We need to approximate now 6 conditional expectations. We specify the ‘core’ and ‘out’ vectors as follows:

$$X_{t+1}^{core} = \left\{ 1, g_{t+1}, \lambda_t, \{b_t^i\}_{i=1, M, N}, \{(b_t^i)^2\}_{i=1, M, N}, \{g_{t+1} b_t^i\}_{i=1, M, N} \right\}$$

$$X_{t+1}^{out} = \left\{ \{b_t^i \lambda_t\}_{i=1, M, N}, \{(b_t^i)^3\}_{i=1, M, N}, \{b_t^i b_t^k\}_{i, k \in \{1, M, N\}, k \neq i}, \lambda_t^2, \{b_{t-i+1}^i\}_{i=M, N}, \{b_{t-i+1}^i \lambda_{t-i+1}\}_{i=M, N} \right\}$$

Therefore we have 12 variables in the core vector and 14 variables in the out vector.

Table 1 summarises the previous discussion on our choices for  $X^{core}$  and  $X^{out}$ .

### B.1.2 An $R^2$ selection criterion for the elements of $X^{out}$

Once we have chosen the composition of  $X^{core}$  and  $X^{out}$ , we apply the following procedure:

1. In implementing the ‘Condensed PEA’ and calculating the linear combinations, we regress each variable  $X_j^{out}$  on  $X_{-j}^{out}$  and  $X^{core}$  and compute the R-square of the regression,  $R_j^2$ .
2. We find the variable  $k$  with the highest  $R_k^2$ , that is  $k = \arg \max_{j \in \{1, 2, \dots, \text{length}(X^{out})\}} \{R_j^2\}$ . If  $R_k^2 > 0.995$  we set the coefficient  $\alpha_k^1 = 0$ . In other words, we set the coefficient of this variable in the first linear combination (and in all approximations) equal to zero.
3. We repeat Steps 1 and 2 removing the excluded variables from  $X^{out}$  until  $R_k^2 < 0.995$ .
4. We apply the ‘Condensed PEA’ to find the coefficients  $\bar{\gamma}^{i,f}$  and  $\bar{\delta}^{i,f}$ , i.e. the new fixed point in the model, with the first linear combination of the elements of  $X^{out}$  which ‘survive’ Steps 1-3.

<sup>8</sup>In other words  $R_t$  can still be identified through the budget constraint or through the nonlinear transaction costs. Had we allowed the government to repurchase more than once and if the transaction costs were assumed independent of (the vector in the case of many repurchases)  $R$  we would need the control variables  $R$  to be in  $X^{core}$  in order to solve the model.

<sup>9</sup>Moreover, since  $R_t$  is always close to zero introducing it as an independent variable in the core vector leads to convergence problems. We discuss this further below.

Table 1: Variables used in approximations

|             | $X^{core}$                                                                                            | $X^{out}$                                              |                                                                                                                                                              | total |
|-------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             |                                                                                                       | common var.                                            | ad hoc                                                                                                                                                       |       |
| BB          | $1, g_{t+1}, \{b_t^i\}_{i=1,N},$<br>$\lambda_t, \{(b_t^i)^2\}_{i=1,N},$<br>$\{g_{t+1}b_t^i\}_{i=1,N}$ | $\{(b_t^i)^3\}_{i=1,N},$<br>$\lambda_t^2, b_t^N b_t^1$ | $\{b_t^i \lambda_t\}_{i=1,N}, \lambda_t b_{t-1}^N,$<br>$\lambda_{t-1} b_{t-1}^N, \lambda_{t-N+1} b_{t-N+1}^N,$<br>$\lambda_{t-N+2} b_{t-N+1}^N$              | 19    |
| NBB         |                                                                                                       |                                                        | $\{b_t^i \lambda_t\}_{i=1,N},$<br>$\lambda_{t-N+1} b_{t-N+1}^N, b_{t-N+1}^N$                                                                                 | 17    |
| coupons     |                                                                                                       |                                                        | $\{\lambda_{t-i} b_{t-i}^N\}_{i=0}^{N-1},$<br>$\lambda_t b_t^1, \{b_{t-i}^N\}_{i=1}^{N-1}$                                                                   | 30    |
| callables   |                                                                                                       |                                                        | $\{b_t^i \lambda_t\}_{i=1,N}, \lambda_t b_{t-N+m}^N,$<br>$\lambda_{t-N+1} b_{t-N+1}^N, \lambda_{t-N+m} b_{t-N+1}^N$<br>$b_{t-N+m+1}^N$                       | 19    |
| repurchases |                                                                                                       |                                                        | $\{b_t^i \lambda_t\}_{i=1,N}, b_{t-N+1}^N - R_{t-N+2}$<br>$(b_{t-N+1}^N - R_{t-N+2}) \lambda_{t-N+1}$<br>$R_{t-N+2} \lambda_{t-N+2}$                         | 18    |
| 3 bonds     |                                                                                                       |                                                        | $\{b_t^i \lambda_t\}_{i=1,M,N}, \lambda_t^2,$<br>$(b_t^M)^3, b_t^1 b_t^M, b_t^M b_t^N$<br>$\{b_{t-i+1}^i\}_{i=M,N}, \{b_{t-i+1}^i \lambda_{t-i+1}\}_{i=M,N}$ | 26    |

5. When we recover  $\bar{\gamma}^{i,f}$  and  $\bar{\delta}^{i,f}$ , we repeat steps 1-4 to determine which of the variables in  $X^{out}$  have a non-zero coefficient in the second linear combination. We apply this procedure to all linear combinations we include to the model.

To understand why the above criterion is useful notice that when  $R_j^2 > 0.995$ , most of the variability of  $X_j^{out}$  is either explained by the core state variables and/or  $X_j^{out}$  is highly correlated with other variables in  $X^{out}$ . In the first case the residuals of the regression of  $X_j^{out}$  on  $X^{core}$  (required to estimate the linear combination) will be close to zero so that the variable does not add almost anything to the approximation. In the second case, the residuals will be highly correlated with the residuals of other  $X^{out}$  variables. In both cases estimating the coefficients  $\alpha$  becomes problematic and the convergence of the model with linear combinations becomes more difficult. Since a high  $R_k^2$  denotes that the  $k$ -th variable is redundant, it helps the algorithm to converge if its coefficient is set to zero beforehand.

### Number of linear combinations used in the approximation

Tables 2 to 4 summarize the number of linear combinations we add to each approximation to some of the models considered in the paper. Consider first Table 2 which reports the results for the buyback models (under ‘no lending’, top panel and under ‘lending’, bottom panel).

In order to decide whether a new linear combination is needed we study how much of the variation of the expression inside the conditional expectation is explained by the new linear combination. In this subsection,  $R^2$  denotes this coefficient of variation. So, for example, for the Euler equation involving equation (20) in the main paper,  $R^2$  now denotes how much of the variance of  $u_{c,t+i}$  is explained by a certain linear combination of the  $X^{out}$ .

The rows in the table summarise the gains in  $R^2$  in percentage terms for each linear combination.  $R_{aug}^2$  is the value of the coefficient of variation we obtain when we include an additional linear

Table 2: **Linear Combinations: Buyback Model**

|                                                 |        | <i>BuyBack 'no Lending'</i> |             |               |                          |                          |
|-------------------------------------------------|--------|-----------------------------|-------------|---------------|--------------------------|--------------------------|
|                                                 |        | $u_{c,t+1}$                 | $u_{c,t+N}$ | $u_{c,t+N-1}$ | $u_{c,t+1}\lambda_{t+1}$ | $u_{c,t+N}\lambda_{t+1}$ |
| $\frac{R_{aug}^2 - R_{old}^2}{R_{old}^2} * 100$ | $LC_1$ | <b>0.0757</b>               | 0.0169      | <b>0.1677</b> | 0.0441                   | 0.0258                   |
|                                                 | $LC_2$ | 0.0026                      | 0.0228      | 0.0043        | <b>0.0547</b>            | 0.0417                   |
|                                                 | $LC_3$ | 0.0259                      | 0.0232      | 0.0234        | 0.0060                   | 0.0308                   |
|                                                 | Total  | 1                           | 0           | 1             | 1                        | 0                        |
|                                                 |        | <i>BuyBack 'Lending'</i>    |             |               |                          |                          |
|                                                 |        | $u_{c,t+1}$                 | $u_{c,t+N}$ | $u_{c,t+N-1}$ | $u_{c,t+1}\lambda_{t+1}$ | $u_{c,t+N}\lambda_{t+1}$ |
| $\frac{R_{aug}^2 - R_{old}^2}{R_{old}^2} * 100$ | $LC_1$ | 0.0081                      | 0.0451      | 0.0403        | 0.0385                   | 0.0322                   |
|                                                 | Total  | 0                           | 0           | 0             | 0                        | 0                        |

Note: The table shows the number of linear combinations in the buyback models ('no lending', top panel and 'lending', bottom panel). The columns list the conditional expectations we approximate in these models. The rows report the percentage gains in  $R^2$  from adding a further linear combination to the model. Hence row  $LC_1$  shows the gains when we compare the regressions with  $X^{core}$  only ( $R_{old}^2$ ) to the regressions with  $X^{core}$  and one linear combination ( $R_{aug}^2$ ). In row  $LC_2$   $R_{old}^2$  derives from a regression on  $X^{core}$  and the first linear combination and  $R_{aug}^2$  adds a second linear combination and so on.

We denote in bold values of  $\frac{R_{aug}^2 - R_{old}^2}{R_{old}^2} * 100$  which exceed the 0.05 percent threshold (above which we introduce an additional linear combination to the model).

combination to the model.  $R_{old}^2$  the coefficient of variation without the additional linear combination. The row labeled  $LC_1$  corresponds to the 'Condensed PEA' test when we solve the model only with the  $X^{core}$  variables and test the inclusion of the first linear combination.  $LC_2$  tests the significance of the second linear combination and so on.

We add a further linear combination to an approximation when

$$R_{diff}^2 = \frac{R_{aug}^2 - R_{old}^2}{R_{old}^2} * 100 > 0.05,$$

in other words when the gain in  $R^2$  is greater than 0.05 percent.

As Table 2 shows the buyback model under 'no lending' requires one linear combination. The approximations of  $E_t u_{c,t+1}$  and  $E_t u_{c,t+N-1}$  include a linear combination in the first round and the approximation of  $E_t u_{c,t+1} \lambda_{t+1}$  includes one in the second round. In the buyback 'lending' model the importance of the  $X^{out}$  variables is limited and so this model does not require any linear combinations.

Table 3 reports the analogous findings in the no buyback models and Table 4 for the case of coupons. Each of these models is solved with linear combinations.

Table 3: **Linear Combinations: No-Buyback model**

|                                                 |        | $u_{c,t+1}$                    | $u_{c,t+N}$   | $u_{c,t+1}\lambda_{t+1}$ | $u_{c,t+N}\lambda_{t+N}$ |
|-------------------------------------------------|--------|--------------------------------|---------------|--------------------------|--------------------------|
|                                                 |        | <i>No BuyBack 'No Lending'</i> |               |                          |                          |
| $\frac{R_{aug}^2 - R_{old}^2}{R_{old}^2} * 100$ | $LC_1$ | 0.0173                         | 0.0166        | <b>0.0561</b>            | <b>0.0646</b>            |
|                                                 | $LC_2$ | <b>0.0578</b>                  | 0.0112        | 0.0109                   | 0.0035                   |
|                                                 | $LC_3$ | 0.0002                         | 0.0194        | 0.0058                   | 0.0001                   |
|                                                 | Total  | 1                              | 0             | 1                        | 1                        |
|                                                 |        | <i>No BuyBack 'Lending'</i>    |               |                          |                          |
| $\frac{R_{aug}^2 - R_{old}^2}{R_{old}^2} * 100$ | $LC_1$ | 0.0194                         | <b>0.0590</b> | 0.0342                   | <b>0.1448</b>            |
|                                                 | $LC_2$ | 0.0167                         | 0.0007        | 0.0140                   | 0.0006                   |
|                                                 | Total  | 1                              | 0             | 0                        | 1                        |

Note: The table shows the number of linear combinations in the no buyback models ('no lending', top panel and 'lending, bottom panel). See Table 3 for details.

Table 4: **Linear Combinations: No-Buyback Model with Coupons**

|                                                 |        | $u_{c,t+1}$               | $q_{c,t}$ | $u_{c,t+1}\lambda_{t+1}$ | $q_{\lambda c,t}$ |
|-------------------------------------------------|--------|---------------------------|-----------|--------------------------|-------------------|
|                                                 |        | <i>No BuyBack Coupons</i> |           |                          |                   |
| $\frac{R_{aug}^2 - R_{old}^2}{R_{old}^2} * 100$ | $LC_1$ | 0.0067                    | 0.0154    | 0.0213                   | <b>0.0654</b>     |
|                                                 | $LC_2$ | 0.0048                    | 0.0108    | 0.0077                   | 0.0011            |
|                                                 | Total  | 0                         | 0         | 0                        | 1                 |

Note: The table shows the number of linear combinations in the no buyback model with coupons. See Table 3 for details.  $q_{c,t} \equiv \sum_{j=1}^N \beta^j u_{c,t+j} \kappa + \beta^N u_{c,t+N}$   $q_{\lambda c,t} \equiv \sum_{j=1}^N \beta^j u_{c,t+j} \lambda_{t+j} \kappa + \beta^N u_{c,t+N} \lambda_{t+N}$



The picture is similar when we consider the models not included in the tables: the callable bonds, three maturities and the model with optimal repurchases. The callable bond model needs one linear combination to be added to get accurate solutions. For the three maturities model we find that  $X^{core}$  is sufficient and therefore we do not include any linear combinations.

The optimal repurchase model requires two linear combinations to be accurately solved. The first linear combination is introduced to the approximations of  $E_t u_{c,t+i} \lambda_{t+i}$ ,  $i = 1, N, N - 1$  and  $E_t \xi_{U,t+1}^R$ . The second linear combination is introduced to the approximation of  $E_t u_{c,t+1}$ .

## B.2 Some practical features of the numerical implementation

### B.2.1 Dealing with occasionally binding constraints on debt

As explained in the main text, we impose an upper and lower bound on the issuance of short and long bonds. In a two bond model we have in total four constraints. These constraints are only occasionally binding and in theory we could use the approach explained, for example, in Marcet and Singleton (1999) to deal with them. Suppose that the government can issue only one bond, whatever the maturity. Marcet and Singleton suggest for every period  $t$  first to solve the unconstrained problem and check whether one of debt constraints is violated. If it is violated, the value of the bond is set equal to the value of the debt limit and  $c_t$  and  $\lambda_t$  are recalculated accordingly.

Unfortunately, this cannot be easily applied in the case of more than one maturity because of the number of constraints involved. If one of the constraints is violated when solving the unconstrained problem, we need to verify that forcing the constraint is not going to generate a violation of one of the constraints on the other bond. This problem presents too many cases to be checked one by one and the computational burden increases considerably when an additional maturity is introduced to the model. For this reason we impose the following (quadratic) costs when the bonds violate the limits in the buyback model:

$$C(b_t^i) = \begin{cases} \frac{\phi_1}{2} \left( b_t^i - \frac{\overline{M}_i}{\beta^i} \right)^2 & b_t^i > \frac{\overline{M}_i}{\beta^i} \\ \frac{\phi_1}{2} \left( \frac{M_i}{\beta^i} - b_t^i \right)^2 & b_t^i < \frac{M_i}{\beta^i} \\ 0 & otherwise \end{cases}$$

for  $i = 1, N$ .  $\phi_1$  governs the penalty from deviating from the debt limits  $\frac{M_i}{\beta^i}$  and  $\frac{\overline{M}_i}{\beta^i}$ . We choose a value of  $\phi_1$  equal to unity. Analogous cost functions are used in the no buyback and coupons models, the debt limits have to be adjusted in these cases as described in text.

In the optimal repurchase model we have an additional constraint on the level of repurchases:  $0 \leq R_t \leq b_{t-1}^N$ . In this case we continue to impose  $C(b_t^i)$  for  $i = 1, N$  however we use Lorenzoni and Marcet's approach to deal with this extra constraint. When  $R$  violates a limit (either because  $R_t < 0$  or  $R_t > b_{t-1}^N$ ) we fix the value of  $R_t$  to the constraint and solve the FONC to determine the optimal portfolio and the value of the multipliers,  $\xi_{L,t}^R$  and  $\xi_{U,t}^R$ .

### B.2.2 Initial conditions and samples' size

In order to generate a more precise approximation of the policy functions over the debt space we use PEA with oversampling. We choose 25 different initial conditions for the debt levels  $b_{-1}^1$  and  $b_{-j}^N$ , where  $j = 1, 2, \dots, N-1$  uniformly distributed in the interval  $\left[\frac{\underline{M}_i}{\beta^i}, \frac{\overline{M}_i}{\beta^i}\right]$  (e.g. in the buyback model). We draw 10 samples of 500 periods for each initial condition. The total number of observations is then 125000.

Given the initial conditions for the portfolio, we also need to specify some initial values for the  $\lambda$ 's. For this purpose we recover initial values  $\lambda_{-N} = \dots, \lambda_{-1}$  that would be consistent with the deterministic steady state. As is well known in steady state the debt level in these models is indeterminate and so we can obtain a different  $\lambda$  (consistent with a different  $c$ ) for each bond vector. Under no buyback we obviously need to set  $b_{-1}^N = b_{-2}^N = \dots = b_{-N+1}^N$  to be in steady state.<sup>10</sup>

### B.2.3 Rescaling

To improve the stability of the algorithm, we rescale the variables which enter in  $X^{core}$  and  $X^{out}$ . For example we use  $\frac{b_t^i}{\overline{M}_i}$  and  $\frac{\lambda_t - \lambda^s}{\lambda^s}$  instead of  $b_t^i$  and  $\lambda_t$ . This is applied to every lag of the independent variables used in the approximation. We also rescaled the dependent variables in the PEA regressions by their steady state values such that their means are close to one in the approximations. For example, we regress  $\frac{u_{c,t+1}}{u_c^s}$  and  $\frac{u_{c,t+1}\lambda_{t+1}}{u_c^s\lambda^s}$  on  $X^{core}$  and the linear combinations, to obtain the approximations of  $E_t\left(\frac{u_{c,t+1}}{u_c^s}\right)$  and  $E_t\left(\frac{u_{c,t+1}\lambda_{t+1}}{u_c^s\lambda^s}\right)$  respectively. The same is done for the other expectations.

Rescaling is useful because some of the coefficients could be very small without it. For example, consider the buy back no lending model;  $b_t^N$  can fluctuate in simulations between 0 and  $\frac{\overline{M}_i}{\beta^i} \approx 117$  and its square between 0 and  $117^2$ . It is obvious that the estimated coefficients of these terms may be close to zero. This makes it difficult to find a reliable convergence criterion for the model.<sup>11</sup> Through rescaling the state variables fluctuate between 0 and 1. This improves significantly the stability of our algorithm (see also Judd et al (2011a)).

### B.2.4 Convergence of PEA - Finding Good initial conditions of the coefficients

Den Haan and Marcet (1990) show that PEA does not guarantee convergence. Convergence is more likely if we use good initial conditions for the coefficients. This is even more necessary in the context

<sup>10</sup>Notice that since sample sizes are sufficiently long (500 observations) our results do not change when we set (for example)  $\lambda_{-1} = \lambda_{-2} = \dots = 0$ .

<sup>11</sup>To see this, denote the coefficient of variable  $b_t^N$  in some approximation by  $\gamma_N^i$ . Let  $\gamma_{N,1}^i$  be the update of this coefficient and  $\gamma_{N,0}^i$  the initial value. If we use a stopping rule

$$(24) \quad \text{Converge if } \frac{|\gamma_{N,1}^i - \gamma_{N,0}^i|}{|\gamma_{N,0}^i|} < \epsilon$$

and  $\gamma_{N,1}^0, \gamma_{N,0}^i \approx 0$ , then the behavior of (24) will be very erratic (both very high and very low values are possible, and this does not tell us much about convergence of the model's quantities). Analogously, if we use the convention

$$(25) \quad \text{Converge if } \frac{|\gamma_{N,1}^i - \gamma_{N,0}^i|}{1 + |\gamma_{N,0}^i|} < \epsilon$$

for some  $\epsilon$ , then the algorithm may (wrongly) converge after a few iterations.

In our codes we employ the criterion (25), but since the variables are rescaled, we are sure that coefficients which are small in values, do not matter much for the optimal policy.

of the optimal portfolio under incomplete markets. If the initial coefficients constitute a very poor guess of the equilibrium of the model, then the algorithm may circle for a long time and subsequently diverge.

Good initial conditions for portfolio choice models can be obtained as follows:

1. Solve portfolio models with positive transaction costs.

For example consider solving the Ramsey problem under buyback subject to the following government budget constraint:

$$\sum_{i=1,N} p_t^i b_t^i = \sum_{i=1,N} p_t^{i-1} b_{t-1}^i + g_t - \left(1 - \frac{v_{x,t}}{u_{c,t}}\right) (c_t + g_t) + \sum_{i=1,N} \omega_i (b_t^i)^2$$

where  $\omega_i (b_t^i)^2$  is a transaction cost paid by the government at issuance.<sup>12</sup> It is obvious that in this model the optimal portfolio is determinate (even with the conventional PEA). In the limit when  $\omega_i \rightarrow 0$  we obtain the buyback model considered in this paper, if  $\omega_i \rightarrow \infty$  there is no trade in bonds. Hence, good initial conditions can be found from solving models with positive transactions costs and gradually reducing  $\omega_i$  till 0.

2. Solve models under tight debt limits and gradually loosen them.

We found that models with tight debt constraints converge more easily than models with looser ones. Generally speaking, models with very loose debt constraints can converge to a wrong equilibrium which features for example a constant  $\lambda$  as in the case of complete markets. This holds in particular because running the models with samples of 500 observations may imply that the debt limits are rarely hit, if they are very loose.

To illustrate this property we consider two simple examples under very poor initial conditions for the PEA.

- (a) Suppose that the initial guess for the polynomials is  $E_t u_{c,t+i} = X_{t+1}^{core} \gamma^i = \gamma_0^i + \gamma_1^i E_t g_{t+1} + \gamma_2^i b_t^1 + \gamma_3^i b_t^N + \gamma_3^i \lambda_t$  and  $E_t \lambda_{t+1} u_{c,t+i} = \lambda^* (X_{t+1}^{core} \gamma^i)$ . Assume that under very loose bounds (e.g.  $-\underline{M} = \overline{M} = \infty$ ) for every  $t$  we get  $\lambda_t = \lambda^*$  as a solution to the system of FONC.
- (b) Suppose that  $E_t \lambda_{t+1} u_{c,t+N} = \alpha \lambda^* (X_{t+1}^{core} \gamma^N)$  where  $\alpha < 1$  and  $E_t \lambda_{t+1} u_{c,t+1} = \lambda^* (X_{t+1}^{core} \gamma^N)$ . Suppose further that  $\underline{M} = 0, \overline{M} = \infty$ . For every  $t$  the optimal portfolio features  $b_t^N = 0$ .<sup>13</sup>

Under both (a) and (b) the algorithm will not converge. When we update the coefficients  $\gamma$ , the independent variable  $\lambda_t$  ( $b_t^N$ ) is constant under (a) (b), so that the least squares problem cannot be solved. Assuming tight bounds can mitigate this problem. To see this note that when a bound is hit under a) the Euler equations do not hold. The value of  $\lambda_t$  is recovered from the FONC of consumption and generally it will be that  $\lambda_t \neq \lambda^*$ . This introduces variability in  $\lambda_t$ . Similarly under (b), when  $b_t^1$  reaches its upper bound we will have  $b_t^N > 0$ , so that  $b_t^N$  is not constant and equal to zero for every  $t$ .

<sup>12</sup>Since this cost does not distort the household's program, it does not show up in bond prices.

<sup>13</sup>Since the Euler equations cannot hold simultaneously, there are two possible solutions. Either  $\lambda_t = \lambda^s$  or  $\lambda_t = \alpha \lambda^s < \lambda^s$ . But the latter case (which would give  $b_t^N > 0$ ) can be ruled out since  $b_t^1$  must be at the upper bound which is assumed arbitrarily loose here.

Note that though the above examples are extreme, they are not completely unlikely. It is not uncommon to initiate coefficients with steady state values, or choose values so that one of the bonds is always at the lower bound. When the bounds are loose, poor initial conditions may make the PEA circle or diverge. Assuming tight bounds helps the algorithm converging. The converged coefficients can then be used as initial conditions for models with looser bounds and so on.

A similar argument, showing the importance of ‘moving bounds’ for convergence in the PEA, was made by Maliar and Maliar (2003).

### B.2.5 Calculating the sample moments

As discussed in the text, to compute the moments reported in Table 4 in the paper, we simulated the model 1000 times using as initial conditions the values of  $\mathcal{S}_t$  and the market value of debt, we recovered from the data. In 1955 the share of short debt equaled 39% and the initial debt to GDP ratio was 38% in the CRSP sample.

We then computed the values  $b_{1,-1}$  and  $b_{N,-j}$ ,  $j = 1, 2, \dots, N$  in the deterministic steady state such that the initial share and market value of debt are consistent with these targets. For example in the buyback model we have

$$\frac{\beta b_{1,-1}}{\beta b_{1,-1} + \beta^N b_{N,-1}} = 0.39 \quad \beta b_{1,-1} + \beta^N b_{N,-1} = 0.38 * 70$$

The analogous expressions for the other models are omitted for brevity.

Given the initial conditions for the bonds, we found the initial values of  $c$  and  $\lambda$  to satisfy the FONC of consumption and the government budget constraint in the deterministic steady state.

We then simulated the models and computed the market value of government debt and the share of short bonds. Notice that whereas in the buyback models to construct the market values for short and long bonds it is sufficient to use the approximations of  $E_t u_{c,t+1}$  and  $E_t u_{c,t+N}$ , in the no buyback model this is not the case. In particular we need to compute the value of non-matured debt in period  $t$ . This requires all the prices  $p_t^j$  for  $j = 2, 3, \dots, N - 1$ . Since these prices do not affect the equilibrium properties of optimal allocations, we computed the approximations through simple regressions of  $u_{c,t+i}$  on  $X_{t+1}$ <sup>14</sup> once our algorithm has converged. In other words, we do not use the ‘Condensed PEA’ to calculate these bond prices since these approximations are performed when the algorithm has converged. The model’s convergence properties and the equilibrium allocation do not depend on them.

Finally, note that because the model is solved with quadratic costs if the debt limits are violated, as described in subsection B.2.1, the market value of government debt can become (slightly) negative in the no lending models, in some periods and samples. The statistics reported in Table 4 in the paper are calculated after dropping samples where the market value becomes negative in ‘no lending

<sup>14</sup>We used all the elements of  $X_t$  not just the ones included in  $X_t^{out}$  above. In other words we have all bonds and cross terms  $b_{N,t-j}$  and  $b_{N,t-j}\lambda_{t-j}$  respectively for  $j = 0, 1, \dots, N - 1$ , in these regressions. The logic is the same as described previously. Since all of these terms will appear in the first order conditions in periods  $t+2, t+3, \dots, t+N-1$  we can claim that they exert a direct impact on the conditional expectations. To simplify we used the same state vector for all the additional conditional expectations we wanted to approximate. This is possible because the stability properties of our algorithm do not depend on the extra terms.

models'. For the same reason in order to avoid having a negative share of short debt in simulations (if say  $b_t^1 < 0, b_t^N > 0$ ) or greater than unity (i.e. when  $b_t^1 < 0, b_t^1 > 0$ ), we computed the moments using  $\min\{\max\{\mathcal{S}_t, 0\}, 1\}$  in the no lending models: we forced the share to be equal to zero when it was negative and 1 when it exceeded unity. This adjustment obviously was much more frequent in the buyback 'no lending model' than in the no buyback model.<sup>15</sup>

### B.2.6 Some limitations of the PEA

We cannot claim that the numerical algorithm we propose in this paper can solve every portfolio choice problem under incomplete markets. To make this point, we describe here a few cases where the approximation of conditional expectations under 'Forward States' may not compute accurately equilibria with multiple assets.

The first noteworthy difficulty of our methodology is that as the number of assets increases the optimal allocation may be close to the complete markets' one. Recall that in this case the portfolio and the multiplier  $\lambda$  are constant through time. Clearly, such equilibria cannot be approximated with polynomials of the form  $E_t u_{c,t+i} = \gamma_0^i + \gamma_1^i E_t g_{t+1} + \gamma_2^i b_{1,t} + \gamma_3^i b_{N,t} + \gamma_3^i \lambda_t + \dots$ ; if the RHS variables are roughly constant, the estimation of the polynomial coefficients will not be reliable. Our algorithms are designed to deal with cases where markets are incomplete, this involves either few assets, or tight debt constraints or both.

Second, even under incomplete markets if the government can trade three or more maturities we cannot rule out equilibria where some of the assets are roughly constant over time (and thus stable coefficients for the polynomials are hard to obtain). To see this assume that in the *buyback model* the government issues three different maturities: 1 year,  $M$  year and  $N$  year bonds where  $1 < M < N$ . To take advantage of fiscal hedging the government will likely adopt the following debt management strategy: set  $b_t^1 = \frac{M_1}{\beta} < 0$ ,  $b_t^N = \frac{M_N}{\beta} > 0$  and use  $b_t^M$  to finance deficits and surpluses over the business cycle.<sup>16</sup> Therefore the values of  $b_t^1$  and  $b_t^N$  will be roughly constant over time and thus it will be difficult to construct approximations of the conditional expectations when the polynomials contain  $b_t^1$  and  $b_t^N$  as state variables.

Third, (we found that) it is not as easy to solve models where the government issues bonds of maturities close to one another (e.g. 1, 9 and 10 years or 1, 2 and 10 years) as it is to solve models where maturities are further from one another (e.g. 1, 5 and 10 years). As we have seen, no buyback models give us portfolios where all issuances have the tendency to comove strongly, however, when bonds are close substitutes, asset prices also comove strongly and so portfolios are difficult to pin down. The algorithm then tends to circle without converging.

Note that these difficulties are not relevant if the goal is to solve models with multiple assets and realistic frictions (e.g. imperfect substitutability among the assets). Small transaction costs, bond clienteles and preferences for short term (safe) assets, will give well defined demand curves for each maturity and these are realistic features of government debt markets. Our methodology is therefore

<sup>15</sup>Recall that one of the main findings of the paper is that under no buyback long and short debt levels comove strongly. This property also holds for the issuances  $b_t^1$  and  $b_t^N$ . It is therefore rare that  $b_t^1$  is slightly negative and  $b_t^N > 0$ . If this occurs in our simulations it is likely that the overall market value is slightly negative in which case the sample is dropped as described previously.

However, under buyback and no lending, we frequently have  $b_t^N \gg 0$  and  $b_t^1 \approx 0$  so that small negative values of short debt can occur. In these cases we set the short term share to 0.

<sup>16</sup>If the debt limits on  $b_t^M$  are sufficiently loose, this strategy is feasible.

broadly applicable to solve models with many assets, the limitations described in this subsection arise because our model is a simplistic one and abstracts from several realistic frictions.

### B.3 Accuracy of the solutions

To check the accuracy of the solution we run Euler Equation Error (EEE) tests (see for example Arouba et al (2006) for an exhaustive description of the methodology).

The tests require to numerically calculate each of the conditional expectations in the Euler equations, using the approximated policy functions. Since we have expectations up to  $N$  leads, we use Montecarlo integration to approximate the integrals. We simulate our model for 450 periods and for each of the 250 samples (25 initial conditions times 10 samples per initial condition) using our approximation. We discard the first 200 periods of each sample, and for each subsequent period <sup>17</sup>,  $\bar{t}$ , we draw  $k = 10000$  different shock paths of length  $N$ . We solve our model for each shock path separately using our approximated policy functions. For example for the buy back and no buy back models we then compute the numerical expectations as:

$$\begin{aligned}\Xi_{\bar{t},1} &= E_t \left( u_{c,\bar{t}+1} \right) = \frac{\sum_{i=1}^k u_{c,\bar{t}+1}^i}{k} \\ \Xi_{\bar{t},2} &= E_t \left( u_{c,t+N} \right) = \frac{\sum_{i=1}^k u_{c,\bar{t}+N}^i}{k} \\ \Xi_{\bar{t},3} &= E_t \left( u_{c,t+1} \lambda_{t+1} \right) = \frac{\sum_{i=1}^k u_{c,\bar{t}+1}^i \lambda_{\bar{t}+1}^i}{k} \\ \Xi_{\bar{t},4}^{NBB} &= E_t \left( u_{c,t+N} \lambda_{t+N} \right) = \frac{\sum_{i=1}^k u_{c,\bar{t}+N}^i \lambda_{\bar{t}+N}^i}{k} \text{ or } \Xi_{\bar{t},4}^{BB} = E_t (u_{c,t+N} \lambda_{t+N}) = \frac{\sum_{i=1}^k u_{c,\bar{t}+N}^i \lambda_{\bar{t}+N}^i}{k}\end{aligned}$$

Since we have two Euler equations, we checked separately each of them, calculating the value of the multiplier in period  $\bar{t}$  implied by the expectations  $\Xi_{\bar{t}}$ , and given the portfolio  $b_{1,\bar{t}}, b_{N,\bar{t}}$ <sup>18</sup>. We then calculate the implied consumption error using the FONC of  $c_t$ . The results from the accuracy tests are stated (following the relevant literature) in terms of consumption deviations, this enables us to give an economic interpretation to the errors.

In particular we compute the following quantities:

$$\begin{aligned}EEE_{\bar{t}}^1 &= \frac{\tilde{c}_{\bar{t}}^1 - c_{\bar{t}}}{\tilde{c}_{\bar{t}}^1} \\ EEE_{\bar{t}}^N &= \frac{\tilde{c}_{\bar{t}}^N - c_{\bar{t}}}{\tilde{c}_{\bar{t}}^N}\end{aligned}$$

where  $\tilde{c}_{\bar{t}}$  is the consumption implied by the new approximation of the expectations and  $c_{\bar{t}}$  the one implied by our approximation. We compute the average EEE across all samples and initial conditions, the maximum error and the percentage of positive and negative errors. We average over 62500 errors.

As in Arouba et al. (2006) we report the *absolute* errors using base 10 logarithms to make our

<sup>17</sup>For the 3 bond model we check the errors every 5 periods. This choice was made for computational purposes because in this model we have 125 initial conditions and 10 samples per initial condition.

<sup>18</sup>Since the optimal portfolio is determined through 'Forward States' it is not possible to use objects  $\Xi_{\bar{t}}$  to determined new values of  $b_{1,\bar{t}}, b_{N,\bar{t}}$ .

Table 5: **Euler Equation Errors**

|                           |     | <b>BB</b> |            | <b>NBB</b> |            |         |          |          |         |
|---------------------------|-----|-----------|------------|------------|------------|---------|----------|----------|---------|
|                           |     | lending   | no lending | lending    | no lending | coupons | callable | repurch. | 3 bonds |
| <b>EEE</b> <sup>1</sup>   | ave | -3.97     | -3.72      | -3.84      | -3.86      | -3.89   | -3.92    | -3.84    | -3.77   |
|                           | max | -2.30     | -2.28      | -2.50      | -2.64      | -2.60   | -2.51    | -2.79    | -2.32   |
|                           | %+  | 0.41      | 0.51       | 0.48       | 0.44       | 0.43    | 0.44     | 0.48     | 0.51    |
| <b>EEE</b> <sup>N</sup>   | ave | -3.18     | -3.06      | -3.53      | -3.51      | -3.30   | -3.75    | -3.18    | -3.62   |
|                           | max | -1.81     | -1.93      | -2.55      | -2.47      | -1.99   | -2.49    | -2.12    | -2.74   |
|                           | %+  | 0.50      | 0.54       | 0.54       | 0.49       | 0.57    | 0.49     | 0.61     | 0.42    |
| <b>EEE</b> <sup>N-1</sup> |     |           |            |            |            |         |          | -3.23    |         |
|                           |     |           |            |            |            |         |          | -1.94    |         |
|                           |     |           |            |            |            |         |          | 0.40     |         |
| <b>EEE</b> <sup>M</sup>   | ave |           |            |            |            |         |          |          | -3.59   |
|                           | max |           |            |            |            |         |          |          | -2.30   |
|                           | %+  |           |            |            |            |         |          |          | 0.58    |

findings comparable with the rest of the literature. A value of -3 means a 1\$ mistake per 1000\$, a value of -4 a mistake of \$1 per \$10000 and so on. Table 5 reports the results.

Table 5 shows that the average of the errors are between -3 and -4, that the percentage of positive errors is close to 50% and that the maximum errors are not large. Moreover, we found that it is quite unlikely that region of the state space where the maximum error occurs is visited in simulations. These results are well within the range accepted by other authors (e.g. Aruoba et al (2006)). This suggests that the model solutions are accurate.

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